BANNARI AMMAN INSTITUTE OF TECHNOLOGY
An Autonomous Institution Affiliated to Anna University Chennai - Approved by AICTE - Accredited by NAAC with "A+" Grade
SATHYAMANGALAM - 638 401 ERODE DISTRICT TAMIL NADU INDIA
Ph: 04295-226000 / 221289 Fax: 04295-226666 E-mail: stayahead@bitsathy.ac.in Web: www.bitsathy.ac.in

Lecture Notes

22GE004 & BASICS OF ELECTRONICS

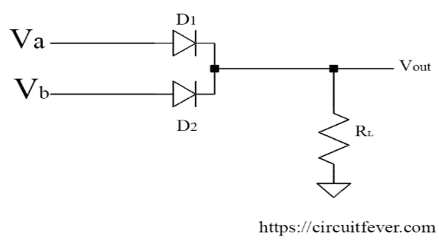
Digital Logic Synthesis using Diode

Logic Gates Using Diodes and Transistor

Logic gates are building blocks of the digital system. In this post, we will see how basic digital gates can be made with the help of diodes and transistor.

OR Gate Using Diode

OR gate has two or more inputs and only one output. The output of the OR gate is HIGH if one or more inputs are HIGH. Circuit diagram of two input, positive logic OR gate using diodes and a resistor is shown below:



In this logic gate circuit, Va and Vb are inputs and Vout is output. These symbols can take only two values either LOW or HIGH.

Working :

Truth table of OR Gate

Va	Vb	Vout
LOW	LOW	LOW
LOW	HIGH	HIGH
HIGH	LOW	HIGH
HIGH	HIGH	HIGH

Let us understand the working of this circuit.

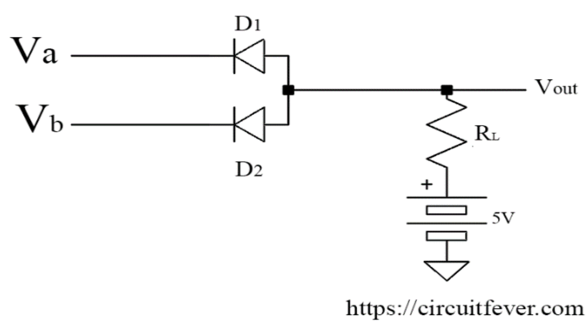
- If all inputs are in LOW, both the diode becomes in reverse biased hence acts as an open switch. Hence the output voltage is LOW.

- If A is HIGH and B is LOW, the diode D1 becomes in forward biased hence act as the closed switch. (Neglecting diode forward resistance and voltage drop across the diode) Hence the output is HIGH.
- If A is LOW and B is HIGH. Diode D2 becomes in forward biased and act as an open switch. Hence the output is HIGH.
- If both the input is in HIGH then the output is equal to the more positive value of the input.
- Hence OR function has been implemented.

AND gate using diodes

AND gate has two or more inputs and only one output. The output of logic AND gate is HIGH if all inputs are HIGH. For other input, the output is LOW.

Circuit diagram of two-input AND gate using diodes and a resistor is shown below:



Here Va and Vb are inputs and Vout is output.

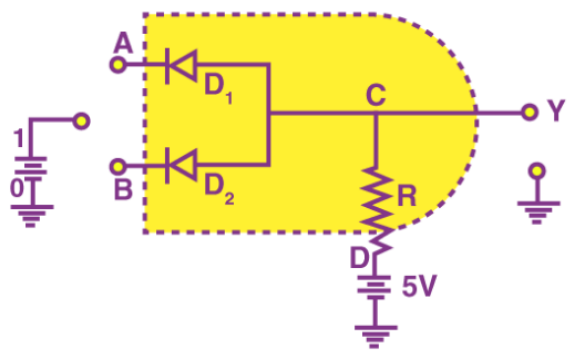
Working :

Truth Table of AND gate:

Input		**Output**
A	**B**	**C**
0	0	0
0	1	0

1	0	0
1	**1**	**1**

An illustrative diagram of a wired-AND connection is as follows:



(i) When both inputs are 1, the two diodes are reverse biased and there is no current flowing to the ground. So the output is 1 as there is no decrease in voltage around the R resistor.

(ii) When one of the logic inputs is ‘0’, the current flows through the corresponding diode and through the resistor. Thus, the ‘0’ logic will be the diode anode or the output.

The working of the circuit can be explained as follows:

Case(i) A = 0 and B = 0.

When A and B are zero, both diodes are in forward bias condition and they conduct and hence the output will be zero because the supply voltage V will be dropped across R only.

Therefore Y = 0.

Case(ii) A = 0 and B = 1.

When A = 0 and B is high, diode D1 is forward biased and diode D2 is reverse biased. The diode D1 will now conduct due to forward biasing.

Therefore, output Y= 0.

Case(iii) A = 1 and B = 0.

In this case, diode D2 will be conducting, and hence the output Y = 0.

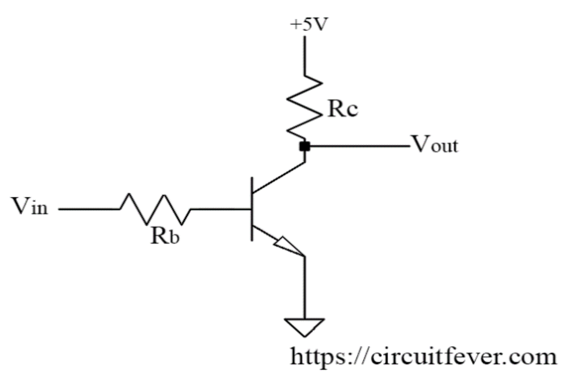
Case(iv) A = 1 and B = 1.

In this case, both the diodes are not conducting. Since D1 and D2 are in OFF condition, no current flows through R. The output is equal to the supply voltage. Therefore Y = 1.

Thus the output will be high only when the inputs A and B are high.

NOT gate using a transistor

NOT gate It is also known as an inverter because the output is opposite to the input. It has one input and one output. Circuit diagram of NOT gate using transistor is shown below:



Here, Vin is input and Vout is output. Output must be LOW if the input is HIGH. Also, the output must be HIGH if the input is LOW.

Working:

Truth table of NOT Gate

Vin	Vout
LOW	HIGH
HIGH	LOW

- If the input is LOW, the parameter is chosen so that the output is V(sat). Also, if the input is HIGH, the parameter is chosen so that the output is LOW.
- If the input is LOW, the transistor act as an open switch. Hence the output is HIGH.

- If the input is HIGH, the transistor act as the closed switch. Hence the output is LOW (Neglecting voltage drop).